

Ce and W co-doped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ with enhanced piezoelectric constant and electrical resistivity at high temperature

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ARTICLE INFO

Article history:

Received 12 September 2019

Received in revised form

17 November 2019

Accepted 9 December 2019

Available online 12 December 2019

Keywords:

Ultra-high temperature ceramics

Piezoelectric properties

Aurivillius phase

Bismuth layered structure

$\text{CaBi}_2\text{Nb}_2\text{O}_9$

ABSTRACT

Ce and W co-doped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ceramics with chemical formula $\text{Ca}_{0.96}\text{Ce}_{0.04}\text{Bi}_2\text{Nb}_{2-x}\text{W}_x\text{O}_9$ (CCBN-W100x, $x = 0-0.07$) are fabricated via conventional solid state sintering method, to investigate the effect of W addition on the structure, electrical resistivity, dielectric and piezoelectric properties. A piezoelectric constant d_{33} of 13.4 pC/N is obtained in CCBN-W2 ceramics, >100% higher than that of pure $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ($d_{33} = 5.8-6.4$ pC/N). Of particular significance is that the electrical resistivity of CCBN-W2 ceramics ($\rho = 3.7 \times 10^9 \Omega \text{ cm}$ at 500 °C) is three orders of magnitude higher than pure $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ($\rho = 2.9 \times 10^6 \Omega \text{ cm}$ at same temperature). All these properties, together with its low dielectric loss ($\tan\delta = 0.13\%$) and excellent d_{33} thermal stability up to 800 °C, merit the CCBN-W2 ceramics for high temperature piezoelectric sensing applications.

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1. Introduction

Calcium bismuth niobate, $\text{CaBi}_2\text{Nb}_2\text{O}_9$ (CBN), is one of the most actively studied compounds among Aurivillius-type bismuth layer-structured ferroelectrics (BLSF) family [1–7]. CBN structure is formed by one unit cell of CaNbO_3 with perovskite-like structure interleaved with fluorite-like $(\text{Bi}_2\text{O}_2)^{2+}$ layers along the c -axis. The ferroelectric to paraelectric phase transition temperature, i.e. Curie temperature T_c , is reported to be ~ 943 °C for CBN. Because of such high T_c , CBN ceramics are good candidates for high-temperature piezoelectric applications [8–10].

BLSF structure promotes a plate-like morphology, with platelets growing preferentially in the fundamental plane (a - b plane) [11,12]. Similar to many other compounds of BLSF family, the properties of

CBN show highly anisotropic [1,3]. Because the spontaneous polarization originates from different rotation modes of $[\text{NbO}_6]$ octahedral, the main component of polarization lies in the a - b plane of perovskite-like CaNbO_3 layer. On the other hand, $(\text{Bi}_2\text{O}_2)^{2+}$ layer, as a natural insulating medium, actually limits the charge transport along the c -axis, leading to the in-plane maximum electrical conductivity in the same a - b plane as the polarization [9]. As a consequence, polycrystalline CBN ceramic sample is normally difficult to pole to achieve full intrinsic performance, accounts for the reported low piezoelectric constant $d_{33} \sim 6$ pC/N [13,14]. Although a very high d_{33} value of 19 pC/N can be obtained in CBN by grain orientation technique, the markedly increased conductivity at high temperature makes it unsuitable for practical application [1,10,15]. Actually, in other textured BLSF ceramics, the resistivity parallel to the texture direction (in a - b plane) has been reported much lower than that perpendicular to the texture direction (along the c -axis) [16,17].

Calcium substitution modification, especially co-modification with alkali metal and rare earth ions, has been extensively studied and reported to effectively improve the piezoelectric properties of CBN ceramics [18,19]. Unfortunately, most of raw materials containing alkali metal ions are soluble in water, and alkali metal

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Peer review under responsibility of The Chinese Ceramic Society.

ions are easy to volatilize during high temperature sintering process, which is not favorable for industrial production. Accordingly, substituting only rare earth for Ca in CBN should be more attractive from the viewpoint of industrial processing. On the other hand, niobium substitution with W or Mn in CBN is also reported to have enhanced piezoelectric properties, owing to donor doping or complex valences to trap the holes originated from charge defect of bismuth vacancy [20,21]. Nevertheless, the high temperature electrical resistivity of CBN ceramics has not been significantly improved, regardless of the modification of Ca or Nb in CBN. Therefore, in the present work, we chose Ce and W to co-modify CBN on the basis of the optimized composition in our recent work [22], while focusing on the effect of W doping content on the structure and its relationship with electrical resistivity and piezoelectric properties, with the target to develop high temperature piezoelectric ceramics with both enhanced piezoelectric constant and electrical resistivity.

2. Experimental

Cerium and tungsten co-doped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ceramics with chemical formula $\text{Ca}_{0.96}\text{Ce}_{0.04}\text{Bi}_2\text{Nb}_{2-x}\text{W}_x\text{O}_9$ (CCBN-W100x, $x = 0, 0.01, 0.02, 0.03, 0.05, \text{ and } 0.07$) were fabricated by conventional processing route using raw materials of 99% CaCO_3 , 99.99% CeO_2 , 99% Bi_2O_3 , 99.99% Nb_2O_5 , and 99.9% WO_3 (Sinopharm Chemical Reagent Co. Ltd., Shanghai, China). The raw materials were weighed according to nominal composition and mixed in a ball mill for 16 h, using a polyethylene jar, zirconia balls and deionized water. The mixture was calcined at 850 °C for 3 h and then ball milled again for 16 h. After drying, the polyvinyl alcohol solution (PVA, 5 wt%) was added in the calcined powders as a binder for granulation. Disks (13 mm in diameter by 0.9 mm thick) were formed by pressing at 120 ± 10 MPa and preheating for 2 h at 650 °C to remove PVA binders, then the disks were sintered for 3 h in an ambient atmosphere with a heating rate of 5 °C/min, the optimized sintering temperature for samples from CCBN-W0 to CCBN-W7 was 1080 °C, 1100 °C, 1120 °C, 1130 °C, 1150 °C, 1160 °C, respectively.

The phase structure of the sintered ceramics was determined by an X-ray diffraction meter (XRD, D8-Advance, Bruker, Germany) using CuK_α radiation ($\lambda = 0.15406$ nm). The XRD profile was analyzed by Jade software. Raman spectroscopy was measured by a Raman spectrometer (LabRAM HR800, HORIBA) at room temperature using laser excitation at 532 nm wavelength to probe the variation of phonon vibration modes. The Raman spectrum was decomposed by Lorentz peak fitting. The microstructural characteristics of polished and thermally etched surface of the sintered ceramics were observed by field-emission scanning electron microscope (FE-SEM, JSM-5610LV, Japan). For electrical measurements, the specimen was polished to about 0.6 mm thick, and then painted with silver paste on both sides and fired for 20 min at 800 °C to form electrodes. Weak-field dielectric constant and loss at 1 kHz were measured by a precision impedance analyzer (HP4294A, Agilent) at room temperature. The electrical resistivity was calculated based on the resistance, which was measured by applying a 100 V to the sample from room temperature to 600 °C using an insulating resistor meter (TH2684) connected to a programmable mini-furnace. The Curie temperature of the ceramics was determined by an LCR meter (TH2828S) connected to a programmable mini-furnace measured from room temperature to 950 °C after sputtering Pt electrode on both sides of the samples. For piezoelectric measurements, the ceramic sample was poled at 180 °C for 20–30 min in silicone oil bath applied with an electric field of 100–150 kV/cm. A quasi-static d_{33} m (ZJ-3A) was used to measure piezoelectric constant d_{33} . In order to evaluate the thermal stability, the poled samples were annealed at different

temperatures for 1 h under short-circuits condition, and then cooled to room temperature for d_{33} measurement. The planar electromechanical coupling factor was determined by resonance method based on IEEE standard on piezoelectricity [23].

3. Results and discussion

As seen from the XRD patterns of CCBN-W100x ceramics shown in Fig. 1(a), all samples are indexed pure orthorhombic phase as reference data (PDF#49-0608) with space group of $A2_1am$. The (115) peak shows the highest intensity, indicating that CCBN-W100x ceramics have bismuth layer structure of $m = 2$, due to the fact that the strongest diffraction peak of BLSF is $(112\ m + 1)$ [24,25]. As can be seen from the enlarged pattern given in Fig. 1(b), (115) diffraction peak slightly shifts to a higher angle with the increase of W^{6+} doping content. This should be owing to the shrinkage of unit cells caused by the substitution of W^{6+} with smaller ionic radius ($r_i = 0.60$ Å, CN = 6) for Nb^{5+} with larger radius ($r_i = 0.64$ Å, CN = 6) [26]. As analyzed by the enlarged pattern given in Fig. 1(c), with the increase of W^{6+} doping content, the characteristic of (200) peak intensity slightly higher than (020) one gradually changes to that of (200) peak intensity slightly lower than (020) one, which indicates that W^{6+} doping should incorporate into the B-site substituting for Nb to alter the rotation and/or distortion of $[\text{NbO}_6]$ octahedron slightly.

Fig. 2(a) shows the Raman patterns of CCBN-W100x ceramics at room temperature. It can be seen that the Raman pattern of all specimens have similar configurations, while the intensity and position of the Raman peaks vary with W^{6+} doping level. In order to determine the effect of W^{6+} doping on the vibration modes, the Raman patterns are decomposed by Lorentz peak fitting, and the fitted pattern for a represented CCBN-W2 sample is shown in Fig. 2(b). As seen from Fig. 2(b), the Raman pattern can be well fitted and decomposed into sixteen independent Raman peaks, which are marked from $\nu 1$ to $\nu 16$ according to the increase of wave number.

Fig. 3 shows the wave number of fitted Raman peaks from $\nu 1$ to $\nu 16$ of CCBN-W100x ceramics as a function of W^{6+} doping content. It was reported that the Raman spectra of CBN-based materials basically show two types of phonon vibration modes: one is high wave number modes appear above 200 cm^{-1} related to vibrations of $[\text{NbO}_6]$ octahedron, and the other is low wave number modes below 200 cm^{-1} related to Ca and/or Bi ions [27–29]. The Raman modes $\nu 8, \nu 9, \nu 15$ and $\nu 16$ are assigned to symmetric metal-oxygen stretching and bending vibrations of $[\text{NbO}_6]$ octahedron, exhibiting minimal change with W^{6+} doping content. For the Raman modes $\nu 6, \nu 10, \nu 12$, and $\nu 14$, the wave number first gradually increases with the increase of W^{6+} doping content, and then drops when W^{6+} doping content x over 0.05, demonstrating that W^{6+} doping can enhance the tilting of $[\text{NbO}_6]$ octahedron. Meanwhile, the decrease of wave number with $x > 0.05$ may be related to the competition of unit cell contraction as shown in Fig. 1(b). The wave number of Raman modes $\nu 11$ and $\nu 13$ first increases and then changes little when W^{6+} doping content $x \geq 0.01$, indicating that W^{6+} addition enhances $[\text{NbO}_6]$ octahedron rotation, but the W^{6+} doping content has no further effect on the rotation of $[\text{NbO}_6]$ octahedron. As a result, the Raman modes $\nu 5$ and $\nu 7$, which comprehensively reflect the distortion degree of $[\text{NbO}_6]$ octahedron in perovskite-like layer, whose wave number can be found to increase a peak value at $x = 0.01$, and then gradually drop with further addition of W^{6+} . The vibration of Ca^{2+} is observed for the Raman modes $\nu 1$ and $\nu 4$, and Bi^{3+} for $\nu 3$, being insensitive to the W^{6+} doping content, which confirms that W^{6+} substitutes for Nb in $[\text{BO}_6]$ octahedron of perovskite-like units, while not Ca^{2+} and Bi^{3+} . The Raman mode $\nu 2$, whose wave number has the same trend as that of modes $\nu 6, \nu 10$,

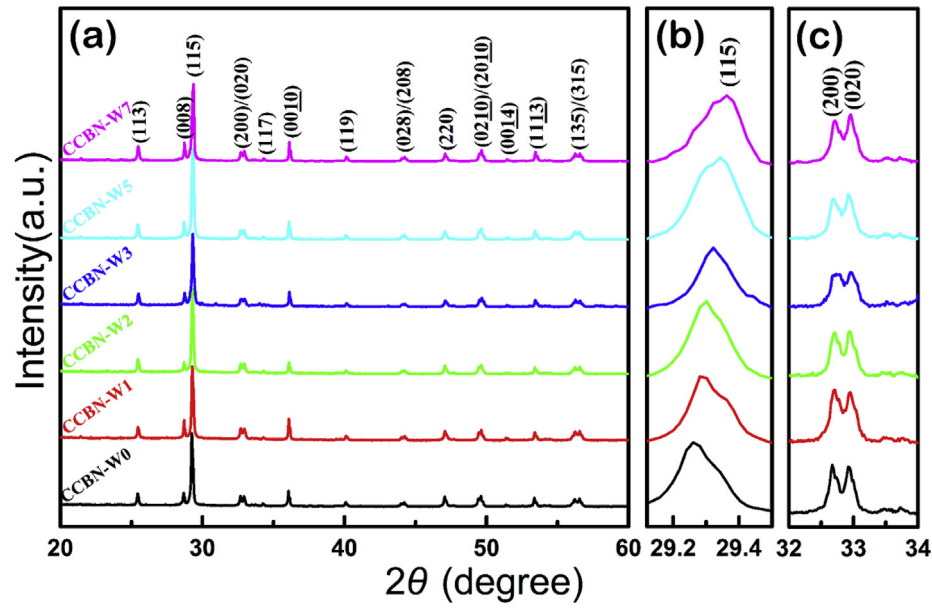


Fig. 1. XRD patterns of CCBN-W100x ceramics.

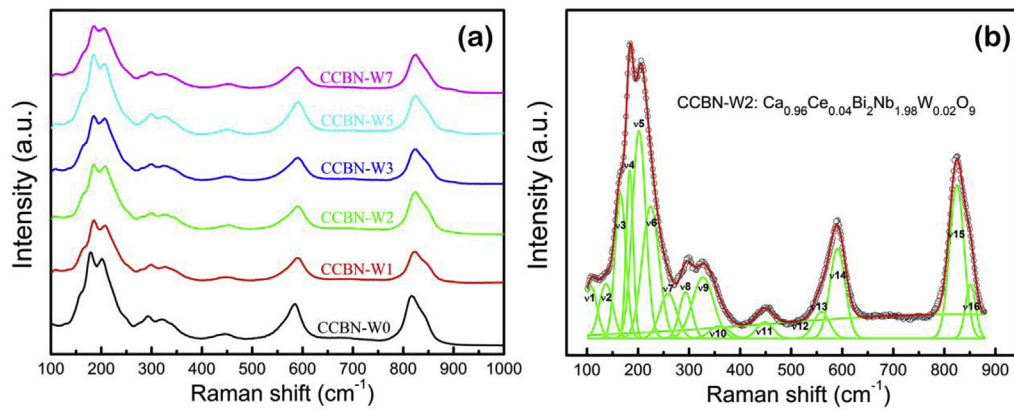
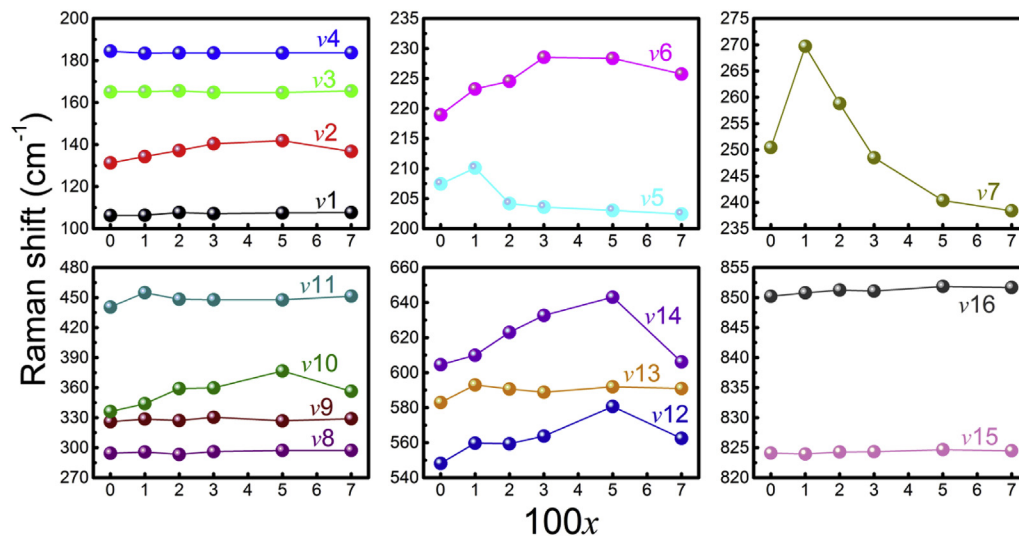


Fig. 2. (a) Raman patterns of CCBN-W100x ceramics at room temperature and (b) Lorentz peak fitting of Raman pattern for CCBN-W2 sample.

Fig. 3. Wave number of Raman peaks from ν_1 to ν_{16} of CCBN-W100x ceramics as a function of W^{6+} doping content.

$\nu 12$, and $\nu 14$ with W^{6+} doping content, should be related to the distortion occurring between $(Bi_2O_2)^{2+}$ layer and perovskite-like $CaNbO_3$ layer [30].

Fig. 4 gives SEM images of thermally-etched surfaces of CCBN-W100x ceramics with different x values. The average grain length and thickness of each sample are shown in Fig. 5. All CCBN-W100x ceramics are found to possess relative density higher than 97.5% as illustrated in the inset of Fig. 5. Generally, BLSF ceramics possess plate-like grains with large aspect ratio, showing anisotropic morphology [3,12,14,20,31]. The small amount of Ce^{3+} doping can effectively inhibit the plate-like grain growth in CBN ceramics, favoring the equiaxed morphology as shown in Fig. 4(a) (CCBN-W0), which should be attributed to low level Ce doping can effectively reduce the ions diffusion along the a - b plane during sintering, thus preventing the plate-like growth of grains as discussed in our previous work [22]. With a small amount W^{6+} addition ($x \leq 0.02$), as observed from Figs. 4(b) and 4(c), the grains still maintain equiaxed growth characteristics. However, typical sheet structured grains are observed in CCBN-W100x ceramics when the W^{6+} doping is above ≥ 0.03 (see Fig. 4(d), 4(e) and 4(f)), analogous to the typical morphology of BLSF ceramics, exhibiting large difference between the length and thickness, as shown in Fig. 5. This phenomenon implies that the introducing of W^{6+} may weaken the equiaxed grains morphology of CCBN-W0 sample by enhancing the ions diffusion along the a - b plane, making the ceramics returns to sheet grain growth again when the W^{6+} doping amount reaches ≥ 0.03 . Meanwhile, it should also be noted that CCBN-W100x ceramics with equiaxed grains ($x \leq 0.02$) have higher relative density than those with plate-like grains ($x \geq 0.03$), as shown in the inset of Fig. 5.

According to the equation of $\sigma = \sigma_0 \exp(-E_a/kT)$, where E_a is the activation energy of charge carrier, k is the Boltzmann constant and T is the absolute temperature, the Arrhenius relationship between $\ln \sigma$ ($\sigma = 1/\rho$) and $1000/T$ of CCBN-W100x ceramics is plotted and

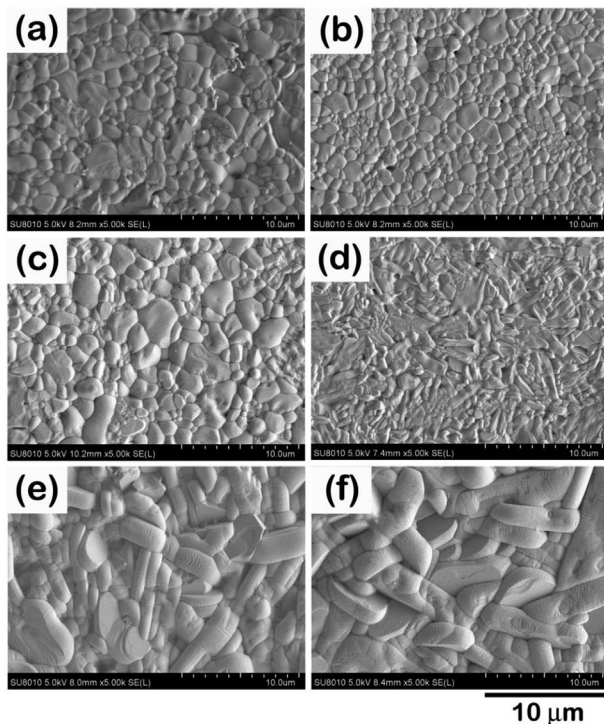


Fig. 4. SEM images of thermally-etched surfaces of CCBN-W100x ceramics: (a) $x = 0$; (b) $x = 0.01$; (c) $x = 0.02$; (d) $x = 0.03$; (e) $x = 0.05$; and (f) $x = 0.07$.

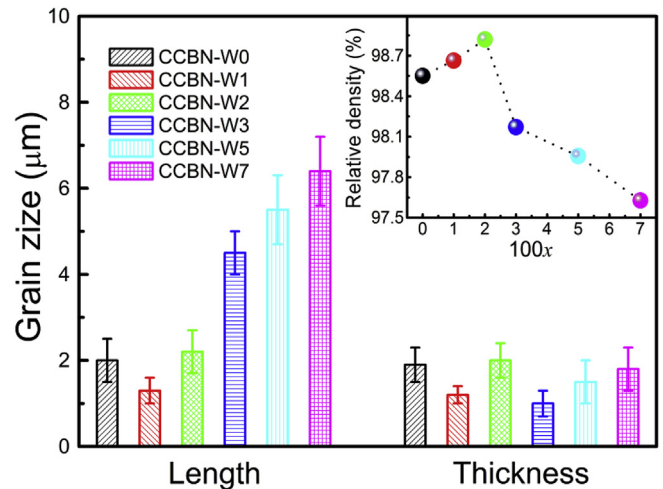


Fig. 5. Grain sizes of CCBN-W100x ceramics. The inset shows the relative density of CCBN-W100x ceramics.

shown in Fig. 6. The conduction behavior of CCBN-W100x ceramics can be characterized by two different temperature regions with an inflexion at 300°C as reported in our previous work [22,24,31,32]. The activation energy E_{a1} , E_{a2} in low and high temperature regions are calculated by the fitted linear slope and given in Table 1. The activation energies E_{a2} in the high temperature region ($>300^\circ\text{C}$) are ranged from 1.51 to 1.92 eV, indicating conduction mechanism is controlled by intrinsic charge carriers. However, the activation energy E_{a1} in the low temperature region ($\leq 300^\circ\text{C}$) is between 0.43 and 0.48 eV, revealing that the low temperature electrical conductivity is dominated by extrinsic semiconductor charge carriers (p -type) [33]. Because bismuth is easily volatilized to form Bi vacancies during high temperature sintering, and Bi vacancy can be ionized to produce holes ($V_{Bi}^{\times} \rightarrow V_{Bi}^{\bullet} + 3h^{\bullet}$), showing p -type conduction mechanism [11,34].

The principle electrical properties of CCBN-W100x ceramics are also given in Table 1 and compared to pure CBN. An enhanced piezoelectric constant ($d_{33} = 13.4$ pC/N) is obtained for the sample with $x = 0.02$ (CCBN-W2), which is $>100\%$ higher than that of pure CBN ($d_{33} = 5.8$ – 6.4 pC/N). In addition to the largest dielectric

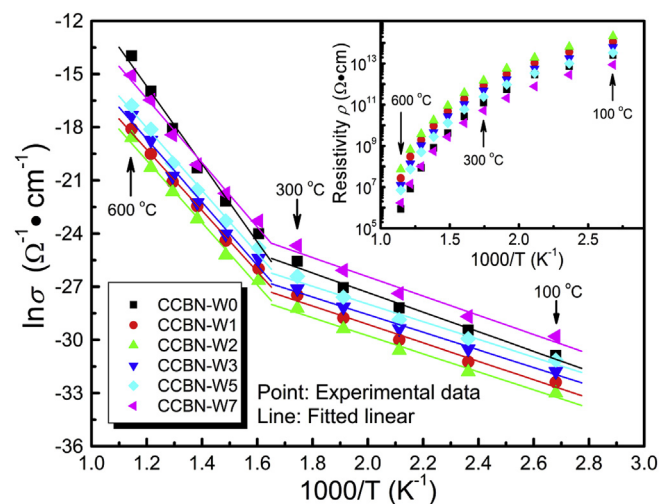


Fig. 6. Arrhenius relationship between conductivity and temperature of CCBN-W100x ceramics. The inset shows electrical resistivity of CCBN-W100x ceramics as a function of temperature.

Table 1

The principal electrical properties of CCBN-W100x ceramics compared with pure CBN.

Sample	$\epsilon_{33}^T/\epsilon_0$ @ 1 kHz	Loss (%) @ 1 kHz	d_{33} (pC/N)	k_p (%)	ρ ($\Omega\cdot\text{cm}$) @ 500 °C	E_{a1} (eV)	E_{a2} (eV)
$x = 0$	106	0.28	10.3	11.5	8.7×10^7	0.48	1.92
$x = 0.01$	109	0.15	11.8	11.8	1.9×10^9	0.45	1.51
$x = 0.02$	111	0.13	13.4	12.2	3.7×10^9	0.44	1.54
$x = 0.03$	108	0.18	11.2	11.7	1.0×10^9	0.43	1.56
$x = 0.05$	107	0.23	10.6	11.2	5.0×10^8	0.43	1.56
$x = 0.07$	102	0.35	9.7	10.6	1.0×10^8	0.47	1.59
CBN ^a	/	/	6.2	/	/	/	/
CBN ^b	90	0.17	5.8	/	/	/	/
CBN ^c	93.5 (@ 100 kHz)	0.24 (@ 100 kHz)	6.4	11.0	2.9×10^6	/	/

^a Data from Ref. [3].^b Data from Ref. [13].^c Data from Ref. [14].

constant and lowest dielectric loss, the CCBN-W2 sample also has the highest electrical resistivity at 500 °C ($\rho = 3.7 \times 10^9 \Omega\text{ cm}$), which is three orders of magnitude higher than that of pure CBN ($\rho = 2.9 \times 10^6 \Omega\text{ cm}$). The greatly enhanced electrical resistivity is thought to be closely associated with the comprehensive effect of the following two aspects. On the one hand, charge compensation caused by donor doping may effectively reduce the concentration of hole charge carriers followed by the two defect eq $3\text{Ce}_{\text{Ca}}^{\bullet} + \text{V}_{\text{Bi}}^{\bullet} \rightarrow \text{null}$ and $3\text{W}_{\text{Nb}}^{\bullet} + \text{V}_{\text{Bi}}^{\bullet} \rightarrow \text{null}$. On the other hand, small equiaxed grains microstructure with the highest relatively density (see Fig. 4(c) and Fig. (5)) will benefit the improvement of resistivity of BLSF polycrystalline ceramics. For Aurivillius-type BLSF materials, as represented by the structural unit of CBN in Fig. 7(a), the charge

transport occurs mainly in a - b plane, while the $(\text{Bi}_2\text{O}_2)^{2+}$ layer along the c -axis acts as a natural insulating medium [1,11,35,36]. Therefore, the activated charge carriers are easy to form a long range migration and then establish macro-conductance behavior inside a - b plane, as a simplified structural schematic of CBN ceramics illustrated in Fig. 7(b), accounts for the lower resistivity in BLSF ceramics with typical plate-like grains. A different scenario, however, is existed in high dense BLSF ceramics with small equiaxed grains as illustrated in Fig. 7(c), i.e., the insulating $(\text{Bi}_2\text{O}_2)^{2+}$ layers and grain boundaries will effectively block the activated charge carrier transportation, making it difficult to form long-range migration and establish macro-conductance, being responsible for the enhanced resistivity. Obviously, the highest electrical resistivity

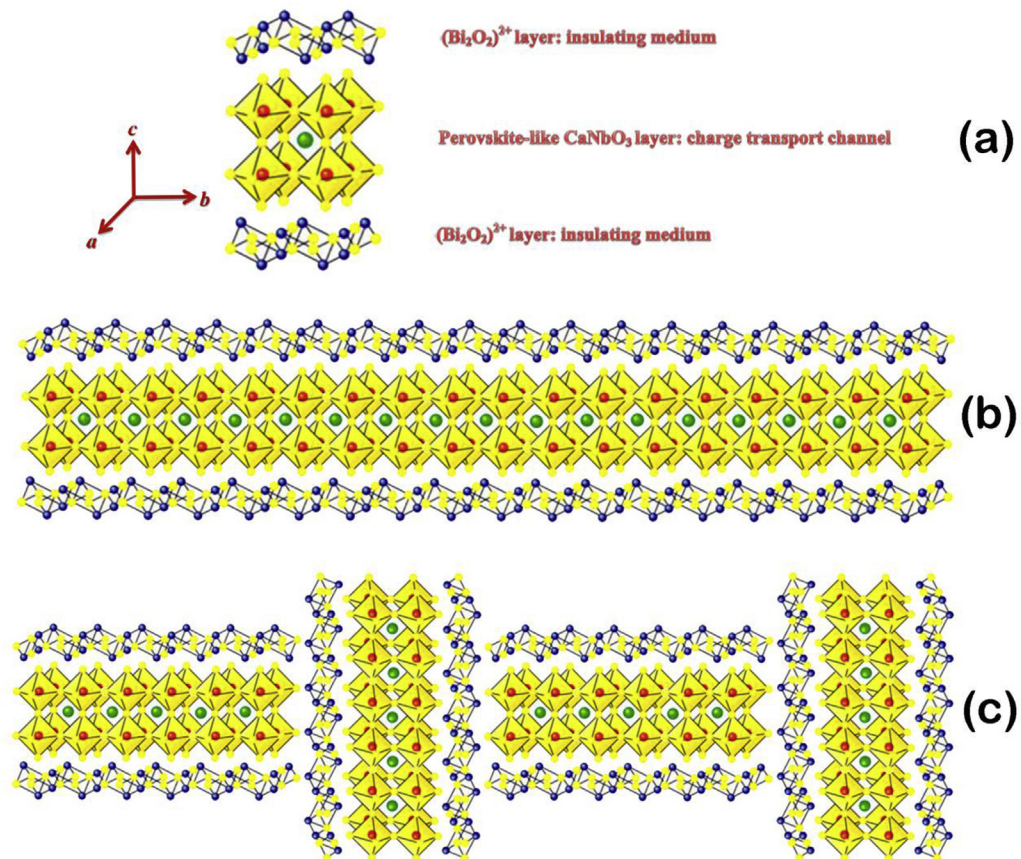


Fig. 7. (a) Structural unit of CBN, (b) Simplified structural schematic of CBN ceramics with large plate-like grains, and (c) Simplified structural schematic of CBN ceramics with small equiaxed grains.

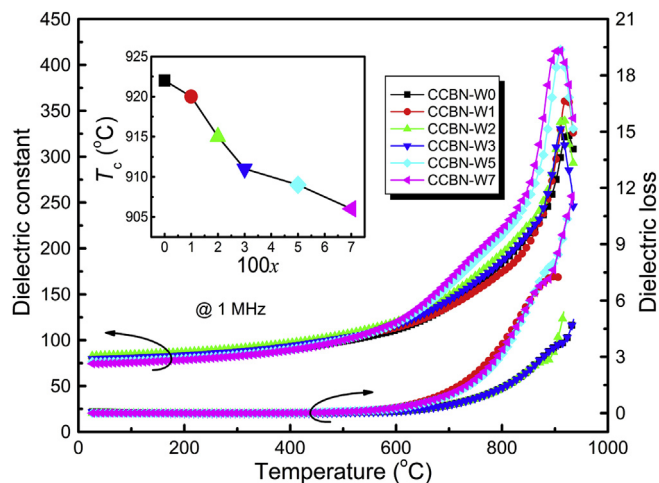


Fig. 8. Temperature dependent dielectric constant and loss of CCBN-W100x ceramics. The inset shows the Curie temperature T_c as a function of W^{6+} doping content.

of CCBN-W2 sample allows higher applicable poling electric field, thus enhancing the polarization efficiency, leading to the improved piezoelectric d_{33} . Actually, our previous work has proved that the more obvious plate-like grains are, the lower the electrical resistivity is, and the deteriorated piezoelectric properties are [31]. Furthermore, it should be noted that the electrical resistivity of CCBN-W2 ceramics ($x = 0.02$) shows the highest value, as observed from the inset in Fig. 6, while the E_{a1} is almost the same for CCBN-W100x ceramics with different x values (see Table 1). This result indicate that the electrical resistivity enhancement in CCBN-W2 ceramic should not due to the change of conduction mechanism, but to the comprehensive effect by the donor doping induced decrease of hole charge carriers concentration and the small equiaxed grains microstructure.

Fig. 8 gives the temperature dependent dielectric constant and loss of CCBN-W100x ceramics, and the inset shows the T_c as a function of W^{6+} doping content. With the increase of W^{6+} doping content, the T_c gradually decreases from 922 °C ($x = 0$) to 906 °C ($x = 0.07$). Fig. 9 gives the temperature dependent dielectric constant and loss of a represented CCBN-W2 ceramics at measuring frequency of 10 kHz, 100 kHz, and 1 MHz. In addition to the Curie temperature T_c , whose location does not vary with the measuring frequency, another dielectric anomaly can be detected around 600 °C–800 °C. With the increasing frequency, the dielectric anomaly peak goes to higher temperature with strongly suppressed dielectric loss, which shows typical dielectric relaxation behavior [37,38]. Such dielectric anomaly should be related to some defect dipoles getting active for polarizing contribution at high temperature and low frequency. The piezoelectric thermal stability of CCBN-W100x ceramics is shown in Fig. 10, and the inset presents room temperature d_{33} of CCBN-W100x ceramics as a function of W^{6+} doping content. With the increase of W^{6+} doping content, room temperature d_{33} first increases and reaches a peak value 13.4 pC/N at $x = 0.02$, and then decreases gradually. All CCBN-W100x ceramic samples exhibit good thermal stability up to 800 °C due to their T_c higher than 900 °C. For example, the d_{33} of CCBN-W2 sample annealed at 800 °C is 12.2 pC/N, retaining 91% of its room temperature value, being promising for high temperature sensing applications.

4. Conclusions

In summary, $Ca_{0.96}Ce_{0.04}Bi_2Nb_{2-x}W_xO_9$ (CCBN-W100x, $x = 0-0.07$) ceramics are fabricated, and the effect of W^{6+} doping content x on structure and properties is investigated. Although W^{6+}

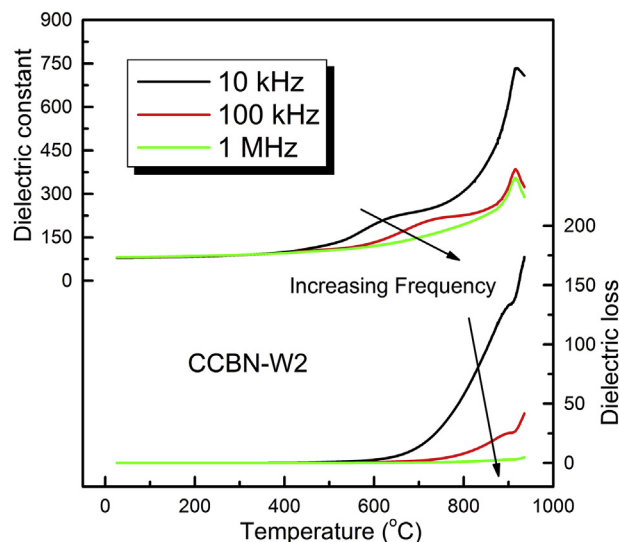


Fig. 9. Temperature dependent dielectric constant and loss of CCBN-W2 ceramics at measuring frequency of 10 kHz, 100 kHz, and 1 MHz.

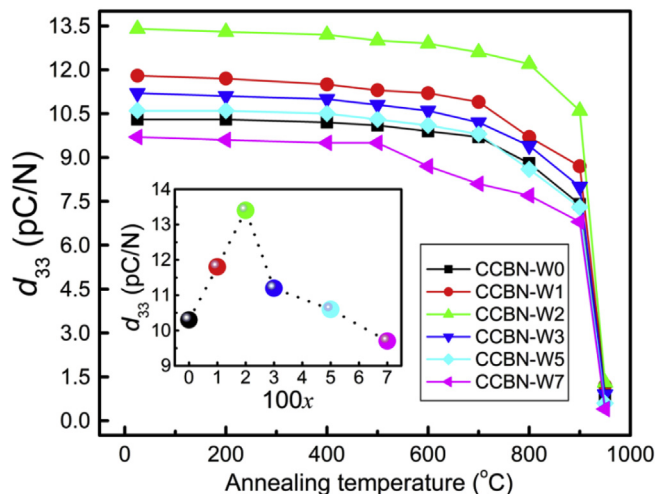


Fig. 10. Thermal stability of the piezoelectric coefficient d_{33} of CCBN-W100x ceramics annealed at various temperatures for 1 h under short-circuits condition. The inset shows room temperature d_{33} of CCBN-W100x ceramics as a function of W^{6+} doping content.

addition reduces the unit cell, it is found to increase the $[NbO_6]$ octahedron tilting and lattice distortion within a certain doping level. Due to the formation of small equiaxed grains microstructure with high density, both enhanced piezoelectric constant ($d_{33} = 13.4$ pC/N) and electrical resistivity ($\rho = 3.7 \times 10^9 \Omega \text{ cm}$ @500 °C) are achieved in CCBN-W2 ceramics. Together with its low dielectric loss ($\tan\delta = 0.13\%$) and excellent d_{33} thermal stability up to 800 °C, the CCBN-W2 ceramic is potential candidate for high temperature piezoelectric sensing applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was financially supported by National Natural Science

Foundation of China (61671224) and Science Foundation of Jiangxi Provincial Education Department of China (GJJ160919). The author Zong-Yang Shen also would like to thank the “Bairen Yuanhang” Project funding supported by Jiangxi Science and Technology Association.

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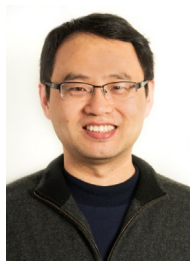
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